LIVE CLASS

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Boats and Streams

Basic Concepts

Tricks & Formulas

Concept Of Upstream and Downstream

Q:1 **A son and father goes for boating in a river upstream . After rowing for 1 mile son notices the hat of his father falling in the river. After 5 min he tells his father that his hat has fallen. So they turn round and are able to pick the hat at the point from where they began boating after 5min. Find the speed of river in miles/hours ?**

A. 4 mile/hr

B. 2 mile/hr

C. 6 mile

D. 5 mile/hr

Q.2 A ship 55 km from the shore springs a leak which admits 2 tonnes of water in 6 minutes, 80 tonnes would suffice to sink her, but the pumps can throw out 12 tonnes an hour. The average rate of sailing that she may just reach the shore as she begins to sink is:

(a) 9.17 km/h (b) 0.97 km/h

(c) 55 km/h (d) 5.5 km/h (e) None of these

Concept Of Upstream and Downstream

TYPE-I

Q3. Speed of a man is 10 km/h in still water. If the rate of current is 3 km/h, then the effective speed of the man upstream is:

Q.4 :

If a man's rate with the current is 12 Km/h and the rate of the current is $1\frac{1}{2}$ Km/h, then his rate against the current is:

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Q. 5 A swimmer covers a distance of 28 Km against the current and 40 Km in the direction of the current. If in each case he takes 4 hours, then the speed of the current is:

(a) 3.5 Km/h

(b) 1.5 Km/h

(c) 2.5 Km/h

(d) 4.5 Km/h

(e) None of these

TYPE-II

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Q6. A boat sails 15 km of a river towards upstream in 5 hrs. How long will it take to cover the same distance downstream, if the speed of current is one-fourth the speed of the boat in still water:

Q7-8 . A boatman can row 2 Km against the stream in 20 minutes and return in 18 minutes. Find the rate of current. In Km/hr

A boat's man goes 48 km downstream in 8 hours and returns back in 12 hours. Find the speed of the boat in still water and the rate of the stream.

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Q9. A boatman goes 2 Km against the current of the stream in 1 h and goes 1 Km along the current in 10 min. How long will he take to go 5 Km in stationary water?

(a) 1 hour

(b) 1 hour 15 minutes

(c) 1 1/2 hours

(d) 40 minutes

(e) None of these

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TYPE-III

Q11: Harsh and vijay move towards hosur starting from iim bangalore at a speed of 40 km/h and 60 km/h respectively. if vijay reaches hosur 200 minutes earlier then harsh what is the distance between iim bangalore and hosur?

a) 600km b) 400km c)900kmd) 200km

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Q.12 A boat travels upstream from B to A and downstream from A to B in 3 h. If the speed of the boat in still water is 9 km/h and the speed of the current is 3 km/h, what distance between A and B?

(a) 6 km

(b) 4 km

(c) 8 km

(d) 12 km

(e) None of these

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Q13: In a river flowing at 2 km/h, a boat travels 32 km upstream and then returns downstream to the starting point. If its speed in still water be 6 km/h, find the total journey time.

(a) 16 hrs
(b) 12 hrs
(c) 14 hrs
(d) 15 hrs
(e) None of these

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Q14: In a stream that is running at 2 km/h, a man goes 10 km upstream and comes back to the starting point in 55 minutes. Find the speed of the man in still water.

(a) 20 km/h

(b) 22 km/h

(c) 24 km/h

(d) 28 km/h

(e) None of these

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Q15 : A motorboat went the river for 14 km and then up the river for 9 km. If took a total of 5 hrs the entire journey. Find the speed of the river flow if the speed of the boat in still water is 5 km/h.

(a) 1 km/h

(b) 1.5 km/h

(c) 2 km/h

(d) 3 km/h

(e) None of these

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Q16: A man can row at 5 km/h in still water. If the river is running at 1 km/h, it takes him 75 minutes to row to a place and back. How far is the place?

(a) 2.5 km

(b) 3 km

(c) 4 km

(d) 5 km

(e) None of these

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Q17 . A motorboat went the river for 14 km and then up the river for 9 km. If took a total of 5 hrs the entire journey.
Find the speed of the river flow if the speed of the boat in still water is 5 km/h.
(a) 1 km/h
(b) 1.5 km/h
(c) 2 km/h
(d) 3 km/h
(e) None of these

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Q18:

A boat can travel at a speed of 15.5 km/h in still water. If speed of the stream is 4.5 km/h, then how much time will it take to go 76 km downstream and 104.5 km upstream?

TITA (_hrs _ Mins) Ex : 5hrs 20Mins

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TYPE-IV

Question 1 :

A boat covers a certain distance against the stream in 9 hours 36 mins and it covers the same distance along the stream in 6 hours. What is the ratio of the speed of the boat in still water to that of the stream?

TITA

Q2:

A boat takes 9 hours to travel a distance upstream and takes 3 hours to travel the same distance downstream.

If the speed of the boat in still water in 4 Km/h, then what is the velocity of the stream?

(a) 4 Km/h

(b) 3 Km/h

(c) 6 Km/h

(d) 2 Km/h

(e) None of these

Question 3 :

A boat running upstream takes 8 hours and 48 mins to cover a certain distance, while it takes 4 hours to cover the same distance running downstream. The speed of current is how much percentage less than the boat in still water?

TITA

Question 4 : Ankit went to Manali, for river rafting. He rows at the rate of 12 km/h in still water. To cover the same distance, it takes him twice as long to row up as to row down the river. Find the rate of flow of the river.

TITA

Q26: The speed of a boat in still water is 50 percent more than its speed in upstream. If the boat takes 20 minutes to reach point Q from point P in downstream, then how much time will it take to reach point P from point Q in upstream?

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Q27: The ratio of the speed of the stream and the speed of a boat in still water is 3: 13. The boat travels upstream for $2\frac{2}{3}$ hours, and then returns to the starting point. How much time (in hours) will it take for the return journey?

TITA

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Q28: The speed of a boat in still water is $\mathbf{5\frac{1}{3}}$ km/h. It is found that the boat takes thrice as much time to row up than it does to row down the same distance in the river stream. Find the speed of the river stream.

TITA

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